Phage therapy for multidrug-resistant *Pseudomonas aeruginosa* infections: a systematic review of the clinical literature and its reporting completeness

Danny Valdiviezo[1], **Nataly Trelles**[1]

[1] School of Medicine, Universidad Católica de Cuenca, Cuenca, Ecuador.

Corresponding author: Danny Valdiviezo. Universidad Católica de Cuenca, Av. de las Américas y Humboldt, Cuenca, Ecuador. Email: dvchiqui[at]gmail.com

Word count: abstract 308; main text 9 764.

Tables: 7. **Figures:** 2. **Supplementary files:** S0–S24 and S18b, plus a guide.

Abstract

Background. Several recent syntheses pool phage therapy outcomes for resistant *Pseudomonas aeruginosa* into overall success proportions. That assumes the evidence base can be aggregated, and no one has examined it.

Objectives. To delimit it reproducibly and determine whether its structure and reporting admit a quantitative synthesis of efficacy.

Methods. PRISMA 2020, separate streams for databases and registers. Eight sources — BVS and SciELO included — over a 2016–2026 window where admitted. English or Spanish only, a criterion adopted **after screening closed** and verified against full text. The unit was the study, with its full article: a criterion added after extraction. Extraction was manual, in duplicate but without established independence, and adjudicated; the structure comes from the model's pre-extraction from abstracts and the outcomes from the adjudicated extraction.

Results. From 23 057 records, 17 129 unique reports remained; 233 reports formed 183 studies and **118 left the corpus**: 63 for lacking the full article —42 existed only as a registry entry— and 55 on the criteria (in 11 they could not be verified) —protocols, preclinical models, and articles where *P. aeruginosa* appears only in the product's declared spectrum—. **65** remain, all read in full text; requiring it left out 7 of the 9 studies with a comparative design. Single case reports, 52.3 %; comparative, 3.1 %; resistance class unassignable, 70.8 %; the language criterion removed 35 studies more. In 43 of the 73 arms (58.9 %) no operational definition of clinical success is on record, and only 2 combine comparative design, numerator, denominator and a definition. One does not separate *P. aeruginosa* and the other measures a time: **none remains**.

Conclusions. This evidence base is at once large and structurally unsuited to a quantitative synthesis of efficacy: published success proportions rest on assumptions these data do not support.

Keywords: bacteriophages; phage therapy; *Pseudomonas aeruginosa*; multidrug resistance; systematic review; PRISMA.

1. Introduction

Pseudomonas aeruginosa adds to its intrinsic resistance an unusual facility for acquiring new determinants. That combination keeps it among the World Health Organization's priority pathogens, though its position has moved: the 2017 list placed it in the critical tier [1] and the 2024 update downgraded the carbapenem-resistant form to high priority [2]. The downgrade does not lessen the clinical problem an isolate with no options poses, but it does mean urgency should not be argued from a superseded classification. Once an isolate accumulates resistance to carbapenems, antipseudomonal cephalosporins, fluoroquinolones and aminoglycosides, the clinician's options can be counted on one hand. Sometimes none is both active and tolerable. The problem concentrates in device-associated infections, extensive burns, cystic fibrosis and bronchiectasis.

Phage therapy has re-emerged into that gap. Lytic bacteriophages act through a mechanism unrelated to antibiotic targets, spare the surrounding microbiota, and can be tailored to a patient's own isolate [3]. But the clinical literature supporting them did not grow the way evidence for a drug in development grows. It grew as an accumulation of rescue cases published one at a time, almost always after the patient improved.

That growth pattern carries consequences that adding more cases does not dissolve. A published case report is, by construction, an event conditioned on its outcome. A compassionate-use series recruits patients in whom antibiotics have already failed and to whom the treating team judged it reasonable to offer something further [4]. And when phage is given alongside active antimicrobials, no favourable result can assign the credit to one or the other [5].

Published syntheses have nonetheless tended to pool this literature into overall clinical success proportions [6,7]. That step requires taking three things on trust that are rarely checked. First, that the resistance category used for stratification survives comparison with the criteria it claims to apply. Second, that the pooled outcome names the same thing across studies. Third, that the intervention is homogeneous enough for a common proportion to mean anything.

A prior question therefore precedes any estimate, and it is the one this review addresses: does this evidence base have the structure, the retrievability and the reporting completeness — that is, the possibility of establishing, from what is published, whether a study is retrievable as a report, what population it concerns and what outcome it measured — that a quantitative synthesis requires? Answering it requires first delimiting the literature completely, including the regional literature and the trial registers that previous reviews have covered unevenly, and then characterising what that body of work states and what it leaves unsaid.

2. Methods

2.1 Protocol and registration

This review was conducted following the PRISMA 2020 statement [8]. The completed checklist accompanies the manuscript as supplementary material. **The review was not registered in PROSPERO or any other prospective registry, and no prior protocol was deposited**; the working protocol and the methodological decisions were date-stamped in the versioned repository from the outset (section 2.8).

2.2 Eligibility criteria

Population. Human patients with *P. aeruginosa* infection classified by the authors as multidrug-resistant (MDR), extensively drug-resistant (XDR) or pandrug-resistant (PDR) according to Magiorakos et al. [9], or described in terms implying those categories. Multi-pathogen studies were eligible when the *P. aeruginosa* subgroup was separable.

Intervention. Therapeutic administration of lytic bacteriophages by any route, alone or combined with antimicrobials. Endolysins and other phage-derived products given without the viral particle were excluded as a pharmacologically distinct intervention.

Comparator. None required, given the expected nature of the literature.

Outcomes. Clinical success, microbiological eradication, mortality, adverse events and emergence of phage resistance, as each study defined them.

Designs. Any design with treated patients: randomised and non-randomised trials, cohorts, case series and case reports. Excluded were laboratory, preclinical and modelling work; reviews without primary data of their own; secondary syntheses; veterinary infections; and studies of other organisms without a separable subgroup.

Language. Reports written in English or Spanish were eligible. **This criterion was not part of the original protocol and was adopted on 11 August 2026, with screening already closed**, so the decision was taken in knowledge of what the corpus contained. That is precisely what prospective registration exists to prevent, and it is therefore declared as an amendment, with its impact measured, in section 2.9 and among the limitations.

Language was not inferred from the journal name, which fails in both directions: *Pirogov Russian Journal of Surgery* publishes in English, and an English masthead can carry an article in another tongue. It was taken from the `language` field declared by Europe PMC, the same datum PubMed files under LA, and, where the source did not index it, from the language field of the export itself, which Scopus, SciELO and BVS all carry. Trial registry entries and Cochrane CENTRAL records were treated as eligible on the published

language rules of those sources rather than on assumption.

Verification did not stop at metadata. For the studies with a retrieved full text the language was checked against the **whole article**, analysing the central pages rather than the front matter, which many journals publish in English even when the body is not. That check exposed three reports whose metadata declared English and whose body is in Russian; all three were excluded. The 233 included reports rest on one of three classes of evidence: analysis of their own text, the language rule of the trial register that publishes them, or the source language field. None is unproven. The check on the body was repeated on 6 October 2026 on the 65 studies of the current corpus with full text, and all 65 are in English. The 3 studies without full text whose language could only be proven on an English abstract, in journals that do not publish natively in English, were excluded (section 3.1.1). The assignment for each report, with its evidence class and proof, is supplied as supplementary material.

Unit of inclusion. The **study**, not the report, following PRISMA 2020. A protocol, its conference abstracts and its final publication constitute one study with several reports. The decision and its justification are recorded in the project decision log. A study is included only if the PDF of its full article is available: the criterion was added on 6 October 2026, with screening and extraction finished, and section 2.9 dates it and measures its effect.

2.3 Information sources and search strategy

Eight databases and registers were searched across the two streams PRISMA 2020 requires to be kept separate. As **bibliographic databases**: PubMed, Scopus (in two arms, one organism-first and one intervention-first), Cochrane CENTRAL via Wiley (issue 7 of 12, July 2026), the BVS portal (sixteen collections, among them LILACS, CUMED, BINACIS and MedCarib) and SciELO. As **trial registers**: ClinicalTrials.gov, EudraCT and CTIS. ProQuest was searched and returned 4 416 records, but was not exported and forms no part of this corpus.

The publication window of 2016 to 2026 was applied in PubMed, in both Scopus arms and in CENTRAL; not in BVS, SciELO or the registers, so 495 records earlier than 2016 entered the corpus. Publication-type restrictions applied in PubMed arm A (six document types), in arm B (human studies) and in Scopus arm B (medicine, immunology and pharmacology). No language restriction was applied. **Embase was not searched**, being unavailable under the institutional subscription, nor was the WHO ICTRP. The documented search runs date from 3 and 4 August 2026, and the BVS export entered the corpus on 10 August; the Scopus run whose export entered the corpus has no recorded date, and the exported file of its arm B finished writing on 5 August.

According to D.V.'s statement (8 October 2026), D.V. designed the search equations with the participation of the language model whose involvement is declared in section 2.4, and ran them in each interface with the model's guidance; the messages of the commits of 3 August 2026, which the model drafted, narrate that execution in the first person. The PubMed and ClinicalTrials.gov records that entered the corpus were downloaded via API with scripts written by the model, and the ClinicalTrials.gov counts were reconciled with those of the interface. The equations are supplied in full as supplementary material, with the literal syntax of each interface and the number of results, so that anyone can repeat them. BVS and SciELO were included so as not to tilt the review towards the anglophone literature, and contributed 2 of the 233 reports that reached full text: they cover Ibero-America, and of the 24 studies that state their origin, 9 are from Russia, Poland, Georgia or Ukraine.

2.4 Study selection

Deduplication matched DOI, PMID and NCT transitively using union-find. Identifiers were read **only** from identifier fields, never from free text. The reason is concrete: in a trial run that scanned abstracts, four unrelated articles were merged because all four cited the same earlier paper. Records without an identifier were compared by normalised title and year, with no fuzzy matching.

Selection proceeded in three stages. **Stage 1** applied explicit, auditable rules: no title, publication type without primary data, animal scope declared in MeSH, and phage used as a laboratory tool. **Stage 2** screened titles and **stage 3** screened abstracts. Both ended up using a **closed vocabulary of six exclusion codes** (ORG, REV, SEC, LAB, VET, OFF), but **it was not fixed before screening began**: the decisions of the first day of title screening, 4 August 2026, carry a free-text reason, and the six codes, whose wording comes from those texts, were used from the following day; that first day's exclusions were later assigned a code. On that same day two rules making the criteria concrete were added mid-batch: isolated endolysins are out on the intervention, and veterinary reports on the population. A reason that fits none of the six does not receive an improvised code: the record advances to the next stage instead of being excluded.

Stage 1 is deterministic: a script runs it and every exclusion names the rule that fired. The record-by-record decisions of stages 2 and 3 were issued by a large language model (Claude Opus 5, Anthropic) as **sole reviewer, without duplication**: 13 917 title decisions and 460 abstract decisions. D.V. was the only human reviewer. According to D.V.'s statement (8 October 2026), D.V. examined one by one the reports that survived screening, but that examination was not recorded. The grouping of those reports into studies was proposed by a script written by the model and is supplied in full (S5). The records the model excluded were not re-read by any human, except the sample of section 2.5. Every decision was logged with its reason and timestamp in an append-only record supplied in full (S3).

Agreement between the model and a human reader was not measured over what the model admitted; over 350 excluded records, the re-screening reader (section 2.5) agreed with the model on 344 in the first pass and on all 350 after checking the criteria. Against the risk of false negatives the review sets the positive-control audit of section 2.5: after every screening stage it was verified that none of the 40 control studies checkable by PMID lost all of its reports.

Two things the word «extraction» conflates are worth separating. The **pre-extraction from abstracts**, which is the source of this article's structural figures (sections 3.1 to 3.4), was issued by the same model under the variable schema fixed in advance, with every record flagged as partial. The **extraction from full text**, from which section 3.5 and Table 5 derive, comes from the workbooks returned by the two reviewers, which the pipeline generated blank. The model intervened around it: it wrote those workbooks, merged into one the two passes returned by the second reviewer, compared the two extractions, drafted the outcome rules —which, according to D.V., both authors decided, although the record of 12 August 2026 left them as an unratified proposal— and translated into categories the corrections the authors signed after re-reading the articles (section 2.6).

Decision logs are **append-only**. A correction is a new row superseding the previous one, and both remain. This review had to correct one erroneous exclusion, and that correction is auditable only because the original row survived.

2.5 Positive-control audit and screening validation

Before screening began, a set of 42 studies known a priori to be eligible was fixed, drawn from the exploratory meta-analysis of July 2026, which was carried out with the assistance of language models; the automated check reached the 40 that had a PMID. The search equations were designed to retrieve that set, so the audit bounds what selection loses, not the sensitivity of the search. After each stage it was verified that **no study in that set lost all of its reports**. The check operates at study level rather than report level, because excluding an erratum or a duplicate registry entry belonging to an included study is legitimate, whereas losing the study itself is not. The audit passed at all three screening stages (40 of 40 checkable) until the language amendment, which excluded Ronit 2024 at the abstract stage (section 2.9). It does not follow the controls after screening: of the 42, 35 remain in the final corpus; besides Ronit 2024, Leitner 2021 left because its full text could not be read, and five for lacking the full article. During one intermediate run it

detected a miscalibrated exclusion rule, which was corrected before proceeding.

Validation over the excluded records. The positive-control audit bounds false negatives only over studies known in advance. To bound them over the unknown ones, a random sample of 350 of the 13 661 records excluded at title or abstract (fixed, reproducible seed) was re-screened in Rayyan by a single reviewer (D.V.), blind to the original verdict and, on the first pass, without the criteria in view. The 6 records D.V. marked as includable proved, when D.V. checked them against the criteria of section 2.2, to be laboratory work, modelling work or review without primary data, and D.V. reclassified them. **The erroneous-exclusion rate was 0.0 % (0 of 350; exact 95 % CI, 0.00 to 1.05 %)**, which over the frame of 13 661 admits up to 143 lost records. The adjudication is supplied as supplementary material.

2.6 Data extraction

The extraction form had 21 data variables, the categorical ones with a controlled vocabulary; no documented pilot is on record. Extraction was carried out **in duplicate** between the two reviewers, but its independence is not established: in the free-text column that locates each datum, the notebook the second reviewer returned on 26 August 2026 matches the first reviewer's character for character in 107 of the 107 comparable rows, with the same typos («MATERILAES Y METODOS», 49 times), whereas on 23 August it matched in none. One reviewer extracted the 124 studies that then had a retrievable publication —65 after the exclusions in section 3.1.1— and the second extracted 122 of them, so 98 % of what was extracted is double-extracted; of the 65 studies in the current corpus, 65 are. Over the 130 comparable arm rows, 541 value disagreements were recorded; agreement, measured before resolving them, was 78 % median agreement and a median Cohen's kappa of 0.62 over the 8 of 9 categorical fields where it is informative, ranging from 64 % to 100 % across variables. For the reason above, that agreement does not correspond to two independent readings and should be read as an upper bound. 539 of 541 disagreements are adjudicated. The adjudication record names a single person in 525 of the 541 rows (293 D.V. and 232 N.T.) and both reviewers in 14: those that had remained open —10 on study design, where the form offered no option the reviewers needed, and 4 for data requiring the article to be re-read—, signed by both on 14 September 2026. The remaining 2 had been closed earlier by a mechanical rule applied to a comparison that was subsequently redone, so they carry no joint signature. The complete record, resolution by resolution, is supplied as supplementary material.

Targeted re-reading after adjudication. Three questions could not be answered with the form and were answered by going back to the article, with one workbook per question in which each conclusion carries the sentence or table it rests on and the signature of both authors, dated 30 September 2026: the resistance class of the 70 studies that then had full text, checked against the published antibiogram (section 3.4, annex S23); the comparator, time zero, assessment time, co-interventions, losses and estimand of the 14 studies with a comparative design, 11 of them with text (section 3.7, annex S24), and the pairs of studies that could describe the same patients (section 4.4, annex S21). The re-reading was done by the two authors, each of whom read the full articles and checked against the figure the antibiograms the article gives only as an image; in addition, every quoted sentence was searched automatically in the text of its PDF. The corrections the re-reading leaves unambiguous enter the extraction as signed corrections, with the previous value and the quotation; those that left an arm with more than one value, and the exclusions the re-reading brought to light, were decided and signed by the two authors on 1 and 3 October 2026.

The structural characterisation of the corpus (sections 3.1 to 3.4, Tables 1 to 4) derives from a systematic pre-extraction from the abstracts of the 95 current studies, carried out under the same variable schema and flagged as partial in every record. Reporting completeness per outcome (section 3.5, Table 5) derives instead from the adjudicated extraction, and is identified as such for every figure. This report presents no efficacy estimates: it characterises the structure of the evidence base, its retrievability and its reporting completeness, with three components measured — retrievability of the report (sections 3.1 and 3.2), structural reporting completeness (section 3.4) and outcome reporting completeness (section 3.5).

2.7 Risk of bias and certainty

Risk of bias is assessed **only in studies with a comparative design**: the 9 that the design adjudicated on the article identifies as comparative, all with full text, as is the whole corpus since the criterion of section 2.9. Randomised trials with RoB 2 [10], and non-randomised trials and cohorts with ROBINS-I [11].

The assessment is issued **at domain level**: one judgement per domain with each instrument's literal categories, without answering the signalling questions through which RoB 2 derives that judgement algorithmically. This is a deliberate simplification in a descriptive review with no pooled estimate, and it is declared so that no algorithmic reproducibility is attributed to it. Case reports and case series are not assessed with a formal instrument, by decision and not by impossibility: they have no comparator, no allocation and no structured follow-up, and a tool built on those assumptions yields a uniform high risk that reports on the instrument rather than on the study [12,13].

There were two notebooks, one in each author's name, but their independence is not established. On the night of 4 to 5 September 2026 both carried every judgement identical and, re-saved two seconds apart, they still shared 7 418 identical characters in their supporting notes; that day's record concludes that one notebook came from the other. The assessment was then declared as by consensus: the pipeline generated a notebook with the matching judgements preloaded and the discrepant ones blank, and both authors signed it on 9 September 2026. Agreement between the two notebooks is reported in section 3.7, but it does not measure two independent assessments. One study, added to the assessment after the others, has a single reading and is declared as such.

GRADE is not applied, and the reason does not depend on the above: it rates the certainty of a pooled estimate and this review presents none. Section 3.6 shows that only 2 of the 73 arms meet the minimum requirements for pooling, and none survives the eligibility criteria. Rating the certainty of a result that does not exist would be a formality, not an assessment [14].

Section 3.7 reports the 68 judgements and the domain table.

2.8 Synthesis methods and reproducibility

The feasibility of a meta-analysis of proportions was assessed explicitly against the usual requirements of that design [15,16]: interpretable denominators, an outcome comparable across studies, and a sufficiently homogeneous intervention. The result of that assessment is presented in section 3.6.

The entire pipeline is written as versioned, re-runnable code: search, deduplication, screening, grouping and computation of every figure quoted. Each record is keyed on an identifier derived from its content and never on its position in a list, because a positional key changes referent as soon as the set of sources changes. An automated check reconciles the output of each stage against the input of the next, and fails if any declared source is not represented in the corpus. During development that check uncovered an entire database that had been searched and downloaded but never reached the corpus.

2.9 Amendments to the protocol

On 11 August 2026, with screening complete and 218 studies already delimited, eligibility was restricted to reports written in English or Spanish. The amendment answers an operational limit of the review team, which cannot extract reliably from a Russian text, and is declared here rather than presented as though it had applied from the outset.

Its effect was measured before it was adopted, and it was not small. It excluded **35 reports** and with them **35 whole studies**, none of which retained a report in an admissible language. The corpus fell from 218 to 183 studies and from 159 to 124 studies with a retrievable publication. The loss was not evenly spread: of 41 comparative designs, 23 remained, and of 21 randomised trials, 16.

Three of those 35 were detected only by reading the article itself. Their metadata declared English; the body is in Russian. The third surfaced only on the publisher's own page, which labels the full text as Russian while the abstract is in English. Had verification stopped at the source language field, all three would have stayed in the review.

The positive-control audit also recorded a concrete casualty. **Ronit et al. (2024)**, a case of phage therapy for a *P. aeruginosa*-infected vascular prosthesis published in Danish in *Ugeskrift for Laeger*, belonged to the set of 42 studies fixed a priori as eligible and is excluded by the criterion. This is not a screening failure, since the study was identified and classified correctly, but the price of the criterion, and it is reported because an

audit that is only reported when it passes demonstrates nothing.

The figures for the review without the amendment survive intact in the decision log, which is append-only, so any reader can reconstruct the earlier corpus and verify the effect independently.

Second amendment: only studies with the full article are included. It was adopted on 2 September 2026 for one study and extended to all on 6 October 2026, both times with screening closed and the second time with extraction finished as well. It is of a different kind from the first, and that should not be glossed over: every other exclusion reason refers to what the **article says** — no patients, no results, the pathogen is on the product label — whereas this one refers to what **this review was able to read**, which is a property of the process, not of the study.

Its measured effect: it excludes **4 studies** under the code NOREC and **63** under the code NOPDF. On 2 September a randomised trial left, of which only a repository copy exists, in a pre-peer-review version and behind an automated-access barrier. On 6 October the two authors excluded under the same code 3 studies whose language, a criterion of section 2.2, could only be checked on an English abstract from journals that do not publish natively in English: EST-045 and EST-059, from a Russian journal, and EST-173, from a Greek one. The same day they signed the extension of the criterion to **every study without the PDF of the full article with access** (code NOPDF): 21 studies whose article was not obtained — 3 of them exist only as conference abstracts — and 42 that exist only as a trial registry entry. The corpus went from 128 to 65 studies.

Until that date this same section advised against generalising NOREC, and the reason still stands: with the extended criterion, full-text retrieval in the corpus is 100 % **by construction**, and the retrieval bias that section 3.2 measured within the corpus can no longer be measured there. It is therefore measured as an **effect of the criterion** (Table 3). Counting both dates, the criterion leaves out 67 studies, and they are not a random sample: it takes **7 of the 9 comparative designs (77.8 %)** and 5 of the 7 randomised trials. The authors adopted it knowing that effect, and it is declared here, with its date, rather than presented as if it had applied from the start.

A third, minor amendment: NOORG. On 14 September 2026 both authors signed a new code for registry entries that declare no organism at all, so the population criterion can be verified neither for nor against; it excluded 7 studies (section 3.1.1). Like NOREC, it refers to what can be verified, not to what the study says. Codes PRO and INT, added on 1 September 2026, do not change the criteria —section 2.2 already required patients treated with bacteriophages— and only name reasons the original vocabulary lacked.

3. Results

3.1 Study selection

The eight sources yielded **23 057 records**, from which 5 928 duplicates were removed to leave **17 129 unique reports** (Figure 1). Stage 1 rules excluded 3 235; of the 13 894 titles screened, 13 434 were excluded and 460 advanced to abstract screening; of those, 227 were excluded and **233 reports proceeded to full text**. Grouped by study, those 233 reports correspond to **183 studies assessed for eligibility**. Of these, **118 left the corpus**: 63 for lacking the PDF of the full article (section 2.9) and 55 on the eligibility criteria: 44 for failing them, checked against the article or against the registry record, and 11 because they could not be verified; the next subsection details them. The evidence base stands at **65 studies**, grouping 80 reports: 60 with a single report and 5 with several, up to a maximum of eleven reports of one trial. The 134 trial-registry reports are not a separate stream from that figure: they are among the 23 057 identified (121 from ClinicalTrials.gov, 10 from CTIS and 3 from EudraCT), and the diagram separates them from screening onwards because PRISMA 2020 requires both streams to be declared.

3.1.1 The 118 studies the screening admitted and that later left

Sixty-three left for lacking the full article (code NOPDF, section 2.9): 21 studies whose article was not obtained — 3 of them exist only as conference abstracts — and 42 that exist only as a trial registry entry, with no locatable associated publication. It is not claimed that they fail the criteria of section 2.2: they were never checked against the article, because this review could not read it. The other 55

left on the criteria: 44 for failing them and 11 because they could not be verified (NOREC and NOORG).

Thirty-eight left on their publication or because it could not be read. They were excluded for eight reasons, and none is a difference of judgement: they are studies that fail what section 2.2 already required. **Thirty** came from judging the article, read in every case except one editorial, judged by its indexing. **Sixteen** have *P. aeruginosa* in no patient — it appears in the declared spectrum of the phage product, in the introduction, in the discussion or in the reference list; one of them names no organism at all. Five of them were detected when the resistance class was checked against the article (section 3.4), and the two authors excluded them on 1 and 3 October 2026. **Five** are study protocols: they describe what will be done and report no outcomes. **Three** have no patients: a murine model of ventilator-associated pneumonia, an *in vitro* hollow-fibre model, and a methods paper on preparing phages in the laboratory. **Two** administer something that is not a bacteriophage: an endolysin, and a product claiming to induce the patient's own native phages. **Two** do not evaluate phage therapy in patients: in one the phage is mentioned only as an unavailable option and was never administered, and the other is a reconstructive surgery article where phage therapy appears once, in a table cell. **Two** supply no primary data of their own: an opinion piece with a narrative review, and a two-page editorial by a single author, which the repository indexing it classifies literally as "Editorial" and "not citable"; its exclusion was signed by D.V. alone.

Four more were excluded on **language**, and were detected differently. All four appeared in the same Russian journal, and the criterion in section 2.2 had kept them on a premise that turned out to be false: that this journal issues a full English version. It does not. Measured on the publisher's own page and **without machine translation**, the body of all four articles is predominantly Cyrillic — between three quarters and almost all of their characters; what is in Latin script is the abstract, the keywords and the references. The site offers no English version by any of its three routes. One of the four also fails on population, as almost all of those above do: its patients had *Staphylococcus* and *Streptococcus*, and *P. aeruginosa* appears only in the composition of the pyobacteriophage administered.

That case exposed a defect in the verification procedure itself, and it is worth declaring. The language detector analyses the title and the **indexed abstract**, yet its strongest evidence class was named "text proven". For a study whose full text was never retrieved that asserts more than was checked, and it applied to **29 of the 95 studies then extractable**. When the source declares a non-admitted language and the abstract is in English, what is proven is that the journal publishes an English abstract — the norm across much of Eastern Europe — not that the article can be read in English. The class was renamed and that precedence corrected.

Four left, finally, under **NOREC** (section 2.9): the full text could not be read and, without it, a criterion could not be verified against the article. One is the randomised trial described in section 2.9, whose exclusion was signed by D.V. alone. The other three — EST-045, EST-059 and EST-173 — are a consequence of that same defect: they only have an English abstract, two in a Russian journal and one in a Greek one that do not publish natively in English, and without the article their language cannot be checked. The two authors excluded them on 6 October 2026.

The cause is common to nearly all of them and is worth stating plainly: title and abstract screening was issued by a language model as sole reviewer (section 2.4). A protocol abstract, a murine-model abstract and one listing *P. aeruginosa* in the spectrum of a polyvalent phage **look very much** like an eligible abstract; telling them apart requires opening the article. Section 2.4 declares that the records the model excluded were never re-read by a human. These 28 are the other side of that limitation — counting those with a stored copy of the text; one article read on the publisher's site and one editorial judged by its indexing have none —, and they are **30.1 %** of the 93 studies whose full text could be read. It is a floor, not a total: the 63 that left for lacking the article were never checked against it.

Another 17 studies left by a different route: their registry record. The earlier 38 have a publication; these have none. They entered the corpus because their exported record carried **the title only**, with no abstract and no MeSH terms, so the screening had nothing with which to rule them out. **29 records** were counted in that situation; the model downloaded the full record of each from ClinicalTrials.gov and from the European CTIS registry, read them one by one and proposed, with its quotation, a verdict for each, which both authors signed. **12** name *P. aeruginosa* among the organisms of the population they will treat and passed the population criterion; as none has a published article, they left later under the full-text criterion (section 2.9). **8** state a different organism in their eligibility criteria — *S. aureus*, *S. epidermidis*, *E. coli*, ESBL-producing Enterobacteriaceae, nontuberculous mycobacteria — and two do not evaluate phage therapy in patients, one being a biobank and the other an observational study whose registered intervention is «none». The clearest case is a phage trial against *E. coli* in children with diarrhoea, whose indexed title — «Antibacterial Treatment Against Diarrhea in Oral Rehydration Solution» — does not let you tell.

The **remaining 7** state no organism at all: their record allows the criterion to be verified neither for nor against. Excluding them required a protocol amendment, because no code in the vocabulary said that.

NOORG is the eleventh, and a sibling of NOREC: the other codes exclude for what the study says, NOREC for what this review could not read, and NOORG for what the registry does not declare. The last two are properties of the process, not of the study. The amendment and the 7 exclusions were signed by both authors on 14 September 2026, and what they cost should be said: these seven studies are part of the evidence for what this work argues, so they leave the total but are counted one by one in Figure 1 and in the exclusions annex. The 29 records travel with the supplementary material, and the literal sentence of the registry supporting each judgement is filed beside them.

Reading them also surfaced a **duplicate**: the Phage4Cure-001 trial is in CTIS under two identifiers, one per protocol version, and had entered as two studies because their titles differ by two words. It is a single trial, verified and signed, and is counted once.

All 65 studies in the corpus have their full article. Before the full-text criterion, **42 of the 128 studies were trial registry entries alone**, with no locatable associated publication. That figure, which the criterion removes from the corpus but not from the field, by itself describes a literature in which almost one registered study in three has never published results.

Exclusion reasons at title screening were laboratory or preclinical work (4 060), other organism without a separable subgroup (3 022), review or commentary without primary data (2 803), does not evaluate phage therapy in patients (2 631), veterinary scope (834) and secondary synthesis (84). At abstract screening: other organism (65), review or commentary (64), does not evaluate phage therapy (40), **inadmissible language (35)**, laboratory (18) and secondary synthesis (5).

3.2 Full-text retrieval and its bias

All 65 studies in the corpus have their full text, obtained through legitimate open-access routes. Since the criterion of section 2.9 that is the entry condition, so the **100.0 %** rate measures the criterion, not retrieval. Retrieval is measured just before it: the text had been obtained for **65 of the 86 studies with an article or abstract (75.6 %)**. The remaining 21 were not obtained. Their availability had been re-checked on 2 September 2026 against three independent indexes — Europe PMC, OpenAlex and Semantic Scholar — and none of the 21 had a locatable open-access copy: some are listed without one and others, the 3 conference abstracts among them, are not listed in the indexes. They were not requested through interlibrary loan or from the authors: the documentary gap is not mitigated by any further route, which is why the criterion excluded them instead of leaving them pending.

The unretrieved fraction **was not a random sample of the whole** (Table 3), and neither is the criterion that excludes it. Counting the 4 NOREC studies, the studies the criterion leaves out contain 7 of the 9 comparative designs, **77.8 %**, although they report fewer patients overall. Of the 7 randomised trials identified with an article or abstract, the corpus keeps 2 — PhagoBurn [17] and BX004-A [18] — and the other 5 left under the criterion, among them the TP-102 trial [19].

The asymmetry is itself a finding. Full-text accessibility covaries with region of publication and with study design, and the full-text criterion pushes in the same direction as the language restriction: both tilt the corpus towards the anglophone open-access literature. A review that declares neither transforms the question without saying so.

3.3 Characteristics of the evidence base

The 65 studies in the corpus were published between **2017 and 2026**, and **86.2 %** since 2020. This is a young and rapidly expanding evidence base (Table 1).

Its composition by design is what a rescue literature produces, not what a synthesis of efficacy needs: **34 single case reports (52.3 %)**, 16 case series, 7 prospective cohorts, 2 randomised trials, 0 non-randomised

trials and 3 retrospective cohorts. A further 3 do not state a recognisable design in the abstract. Comparative designs total **2 studies, 3.1 %** of the whole, and are the only ones able to sustain a claim of relative efficacy.

Where stated, geographic origin is spread across Georgia and Poland (4 studies each), France (3), Belgium and the United States (2 each), and 9 other origins with one study each. It is not stated in 41 of the 65 studies. The fall in Russian representation relative to the pre-amendment corpus, from 34 studies to 1, is mainly an effect of the language criterion, not of the search.

3.4 What the evidence base does not state

Reporting completeness, measured on what abstracts make explicit, is lowest precisely in the variables on which any stratification depends (Table 2):

- **Resistance class** (MDR, XDR or PDR) cannot be assigned in **70.8 %** of studies: 50.8 % do not mention it at all and a further 20.0 % mention it in terms that do not allow classification. For stratifying by resistance, which is what published syntheses do, both cases are equally unusable.
- **Route of administration** is not stated in **49.2 %**.
- **Pathogen scope**, that is, whether the study covers only *P. aeruginosa* or is mixed with a separable subgroup, is not stated in **26.2 %**.
- Fulfilment of the **DTR** criterion (difficult-to-treat resistance) is not stated in **96.9 %**.

These figures describe reporting in the abstract, not necessarily in the full article, and the distinction matters because some of the information may sit in the body of the text. That said, the abstract is what databases index and what automated reviews consume. More than seven studies in ten failing to state their resistance category there means that no MDR/XDR/PDR stratification built from abstracts can be correct. That category is not merely descriptive: it is the review's population criterion. **In 8 of the 73 arms the class falls below the multidrug-resistance threshold** (EST-001 A, EST-003 B, EST-053 A, EST-057 A, EST-088 A, EST-094 A, EST-106 A, EST-108 A). 2 are strata within studies that do include multidrug-resistant patients, and do not compromise the study's eligibility. The other 6 — EST-001, EST-053, EST-057, EST-088, EST-094 and EST-106 — are single-arm studies whose only patient is not multidrug-resistant. EST-001 and EST-094 are kept in the corpus by signed decision of the two authors on 22 September 2026, and EST-053, EST-057 and EST-088, which fell below the threshold when their class was checked against the article (next paragraph), by the same signed decision on 1 October, and EST-106, which fell below the threshold when its class was resolved that same day, by the one signed on 3 October. All are declared here so the reader can discount them.

The class was then checked against the article. The two authors re-read the 70 studies that then had full text, sought the antibiogram and applied the definitions of Magiorakos et al.; each conclusion carries the sentence or table of the article it rests on and the signature of both (annex S23). In 9 the reading found no patient with *P. aeruginosa* treated with phages. The two authors excluded 5 with code ORG (EST-009, EST-055, EST-132, EST-181 and EST-208; section 3.1.1) and kept 4, which are declared here with what the article says: in EST-152 and EST-170 the only mention of *P. aeruginosa* is the description of the patient of another study in the corpus (EST-049 and EST-106), and in EST-129 and EST-135 the patient had *P. aeruginosa*, but the phage was not directed against it. The same holds in EST-051, which is also kept. In EST-096, kept by the same decision, patient 10 did receive a phage against *P. aeruginosa* (Ps1N/734), but the article's two tables contradict each other on whether the bacterium was still present when treatment began. Of the 65 studies with full text that remain in the corpus, the class could be checked in 30: in 22 against a printed antibiogram and in 8 against a description of susceptibility in the text; in 31 there is nothing to check it against — the antibiogram is not published or sits in supplementary material that was not obtained — and the remaining 4 are those kept without treated *P. aeruginosa*. Of the 30 checked, the class matched the coded one in 18; in 5 the extraction had it as not classifiable and the article allowed it to be assigned — 3 below the threshold and 2 XDR —; in 2 it was more severe than coded (EST-043 and EST-047); in 3 the declared class was not confirmed (EST-015, EST-030 and EST-042), and in 2 it changes between patients or between isolates of the same arm (EST-124 and EST-169). The 35 corrections the reading leaves unambiguous for the whole arm — 8 of class, 17 of its source and 10 of the DTR criterion — entered the extraction with the signature of both authors. The 9 studies in which it left an arm with more than one value were resolved by the two authors on 1 October 2026: in 5 the extraction was corrected (EST-047, EST-048, EST-058, EST-106 and EST-124) and in 4 it was left as it was (EST-004, EST-019, EST-042 and EST-169). Two of those four are declared expressly:

EST-042 keeps the PDR its authors declare although its own antibiogram does not support it, and EST-169 keeps in a single arm one patient below the threshold and another XDR.

3.5 What the corpus reports for each outcome

Duplicate extraction, now adjudicated, makes it possible for the first time to measure what this evidence base reports for the five outcomes declared in the eligibility criteria. Table 5 summarises it over the 73 extracted arms.

Completeness is uneven and, at best, partial. Adverse events are the outcome most often reported with a denominator (89.0 % of arms) and emergence of phage resistance the least (35.6 %), despite being the outcome that determines whether the intervention retains its usefulness. The 73 arms belong to studies whose full article was read (section 2.9), so an outcome that is not on record is an outcome the article does not report, not a documentary gap of this review. Clinical success, the outcome that published syntheses pool, is reported with numerator and denominator in 87.7 % of arms. Two arms used to report a numerator larger than their denominator — thirteen clinical successes in one patient and five adverse events in four. On re-reading their articles these turned out to be the numerator of the whole series (13 of 15 patients) and the number of treatment courses rather than of events; both fields were set to NA by signed decision of the two authors on 22 September 2026, and **no arm now reports an impossible proportion**. The denominator for EST-021 is that of its phage arm (13); the article reports clinical success over the mITT population of that arm, which is 12, and the extraction form has no denominator of its own for the outcome, so the pair is recorded as 6/13 and the difference is declared here.

That 87.7 % is misleading, however, if read as availability for pooling. **In 43 of the 73 arms (58.9 %) no operational definition of clinical success is on record**, and in all 43 the reviewers established this by reading the article: 42 state a number of patients with a "good outcome" without saying what had to be observed to count one, and 1 left the field empty. A numerator without a definition names nothing comparable to another study's numerator.

Study design is on record in all 73 arms: **4 arms are trials** (randomised or not) and **8 are cohorts**. Section 3.3 and Table 1 call only the trials comparative, with the design the abstract states, which is why the figure there is 2 studies; when this section says comparative it also includes the cohorts, with the design adjudicated on the article, and there are 12 arms. They are not the same set.

Crossing the requirements a pooled proportion would demand empties the corpus step by step. Four of them are arithmetic — comparative design, numerator, denominator and an operational definition — and leave **2 of the 73 arms**. The next two are the eligibility criteria this review itself declares in section 2.2, and only one of those 2 survives them.

The first is that the outcome be attributable to *P. aeruginosa*. Section 2.2 admits multi-pathogen studies only when the subgroup is separable, and in one of the two the extraction records that the article does not separate the data: its numerator counts patients of several species. One remains. The second is that administration be therapeutic, and that arm's is.

That last one is PhagoBurn, whose primary outcome is the *time* to a sustained reduction in bacterial burden [17]: not a proportion, and one that would enter a meta-analysis of proportions only if the synthesist converted it into one. **None remains.**

Put another way: of the 73 arms of this literature, **not one meets at once the arithmetic conditions, the eligibility criteria and the metric that would make its clinical success a legitimate term in a pooled proportion**. It is not that the available studies are few: there are none. Table 6 sets out the funnel, step by step.

3.6 Feasibility of quantitative synthesis

A meta-analysis of proportions requires interpretable denominators, an outcome that names the same thing across studies, and a sufficiently homogeneous intervention. The body of evidence delimited here fails all three requirements.

Denominators. More than half the studies (52.3 %) are single cases, in which the success proportion can only take the values 0 or 1 and its binomial variance is maximal. Pooling 38 denominators of one with a handful of larger series produces a weighted mean whose weights are set by series size rather than by evidential quality.

Outcome. In the articles of the corpus, "clinical success" neither names the same thing nor measures on the same scale. At one end, composite long-term clinical cure: no implant loosening and walking without pain at 21 months [20]. At the other, discharge criteria: patient discharged, de-cannulated and rehabilitated [4]. In between, the label without a criterion: "good clinical outcome" declared by the authors as binary, without stating what had to be observed [21].

The problem is not only heterogeneity. **The 2 randomised trials in the corpus report no binary success proportion at all**: the primary endpoint of PhagoBurn is the *time* to a sustained reduction in bacterial burden [17], and that of BX004-A is a *continuous* variable, the mean log-scale CFU reduction at day 15 [18]. A meta-analysis of proportions cannot include them without binarising what their authors measured otherwise, and that conversion would be made by the synthesiser, not by the study.

These definitions come from the full texts, not from the pre-extraction from abstracts: **no abstract in the corpus states a definition of clinical success**, which is itself a datum about how the field reports.

Intervention. The products range from fixed-composition commercial polyvalent preparations to cocktails tailored to a patient's own isolate, and in most cases they are co-administered with active antimicrobials. The heterogeneity is conceptual rather than statistical, and no random-effects model resolves it: averaging different interventions yields a number that corresponds to no real intervention.

Added to this is the effect of the full-text criterion documented in section 3.2: 7 of the 9 comparative designs the corpus could have contributed to a synthesis were left out of it. **For these reasons no pooled efficacy proportion is presented here**, and we argue that those published in previous reviews rest on assumptions these data do not verify.

3.7 Risk of bias in the comparative studies

Of the 2 studies assessed with RoB 2, the overall judgement is low risk in 1 and some concerns in 1. Of the 7 studies assessed with ROBINS-I, the overall judgement is no information in 1 and serious risk in 6. The domain concentrating most unfavourable judgements —high risk under RoB 2; serious or critical under ROBINS-I— in ROBINS-I is “bias due to confounding”, in 6 of 7. The two reviewers' notebooks, in 8 of the 9 assessable studies, agree on 49 of the 60 comparable judgements (81.7 %; Cohen's kappa 0.74), an agreement that does not measure two independent assessments (section 2.7). The 11 disagreements were resolved by consensus, all 11 towards the judgement in D.V.'s notebook, none towards the other reviewer's. The remaining study was assessed in a separate workbook and has a single reading.

A comparative design label does not guarantee a comparison group: of the 9 assessable studies, 3 have one (EST-008, EST-021 and EST-108) and in 6 the signed domain-1 notes state there is none. Of the 3 with a comparison group, 2 compare phage with no phage —EST-108 compares with and without antibiotic within phage-treated patients—, and both give a between-arm contrast for *P. aeruginosa* (EST-008 and EST-021; that of EST-008 is partial).

Table 7. Domain-level risk of bias for the 9 assessable comparative studies. RoB 2 for randomised trials (five domains) and ROBINS-I for non-randomised trials and cohorts (seven). “n. a.” marks domains the instrument does not contemplate, not a missing judgement.

4. Discussion

4.1 Principal findings

This review delimits the clinical evidence base for phage therapy in resistant *P. aeruginosa* more broadly than previous syntheses: eight sources, no language restriction, regional literature and trial registers included. The result is 65 studies with their full article; the search had delimited 218 before the language restriction adopted at the end, and 118 left later, 63 of them for lacking the article. That breadth does not speak well of the field's maturity: most of what was identified comes from case reports, from registry entries without publication, and from regional literature that searches confined to PubMed and Scopus do not reach.

The central result is negative and worth stating plainly: **the breadth of this evidence base does not translate into capacity for synthesis**. More than half the studies (52.3 %) are a single case. Only 2 of the 65 are comparative according to the abstract. In more than seven out of ten, the resistance category by which stratification is attempted cannot be assigned. And, before the full-text criterion, almost one study in three (42 of 128) existed only as a registry entry, with no published results.

4.2 Comparison with previous literature

Recent reviews have reported overall clinical success proportions consistently above 80 % [6,7]. Those results do not contradict ours; they explain them. A literature composed largely of cases published after a favourable outcome yields, when pooled, a high proportion by construction. What that proportion measures is the propensity to publish successes, not the efficacy of the intervention.

Our delimitation further suggests that those syntheses worked on a fraction of the field. The unrestricted search retrieved 34 Russian and 3 Ukrainian studies, and that is precisely where comparative designs concentrate. The language amendment obliges us to set them aside, so this review documents that body of literature and where it sits but cannot speak to its content. That is an explicit debt to the next review with translation capacity. The full-text criterion pushes in the same direction: the studies whose article was not obtained concentrate the comparative designs (section 3.2).

4.3 Implications

For clinical practice. No published success proportion for phage therapy in resistant *P. aeruginosa* should be used to estimate the probability of benefit for an individual patient. The available evidence supports the statements that phage therapy is being given, that it is tolerated, and that favourable outcomes have been described. It does not support an effect size.

For research. Three shortcomings could be remedied at no added cost and would transform the usefulness of the next reports. State the resistance category together with the explicit criteria used to assign it. Document that the administered phage had verified lytic activity against the patient's isolate. And define clinical success with an explicit time horizon. The first is the most urgent: without it, the population of the question is undefined.

For registries. That 42 of the 128 studies that reached the full-text criterion were registry entries alone, without a locatable publication, is a first-order publication-bias problem, and no statistical method resolves it; the criterion removes them from this corpus, not from the field.

4.4 Strengths and limitations

Strengths. The search covered eight sources across two separate streams —Scopus in two arms—, with no language restriction. Selection was audited against a positive-control set fixed before screening began: screening lost none of the 40 checkable by PMID until the language amendment, which took one and is declared (section 2.9). The whole pipeline is re-runnable code with content-derived keys, and an automated check reconciles each link with the next.

Limitations. There are eight, and all are material. **First**, the record-by-record decisions of title and abstract screening were issued by a language model as sole reviewer, without duplication (section 2.4), with a vocabulary of reasons that was not fixed before screening began; nobody re-read what it excluded except a sample of 350 records, re-screened by a single reviewer, and agreement between the model and a human reader was not measured over what the model admitted. The positive-control audit and the re-screen bound false negatives without ruling them out: over the frame of 13 661 exclusions, the re-screen admits up to 143 lost records. **Second**, the corpus admits only studies with the full article (section 2.9), a criterion extended to all on 6 October 2026, with screening and extraction finished, which excluded 63 studies, 42 of them registry entries without publication. It depends on the reviewer's access rather than on the study's content, and it is not neutral: it takes 7 of the 9 comparative designs and 5 of the 7 randomised trials. With it, full-text

retrieval is 100 % by construction and retrieval bias is no longer measured within the corpus; it is declared as an effect of the criterion (Table 3). **Third**, the search did not query Embase or the WHO ICTRP, which reduces coverage of case series and conference abstracts and leaves the register stream limited to the United States and the European Union. **Fourth**, it was limited to publications from 2016 to 2026, so it describes recent output rather than the whole field. **Fifth**, this report describes how completely outcomes are reported (section 3.5) but presents no efficacy estimates and no GRADE certainty (section 2.7): it is a characterisation of the evidence base, not a synthesis of its results. Risk of bias is assessed only in the comparative studies with full text, at domain level and by consensus, so it inherits the effect of the full-text criterion. Neither extraction nor risk of bias has established independence between its two readings: the extraction notebooks share a free-text column character for character (section 2.6), and the risk-of-bias notebooks 7 418 characters of notes (section 2.7). Agreement between the two risk-of-bias notebooks was 81.7 % (49 of 60 judgements; Cohen's kappa 0.74), and the 11 disagreements were all resolved towards D.V.'s notebook. **Sixth**, before that criterion the full text had been obtained for 75.6 % of the studies with an article or abstract (65 of 86), and the missing fraction, which the criterion left out, is enriched in comparative designs, so what is described under-represents the comparative part of the field. **Seventh**, the restriction to English and Spanish was adopted once screening had closed and removed 35 studies, 18 of them comparative and one from the positive-control set; it pushes the corpus in the same direction as the retrieval bias and narrows what this review can say about the literature of Eastern Europe. **Eighth**, the corpus counts reports and studies, not distinct patients, and contains **20 patients described in more than one study**, spread over 18 of the 65. Three groups concentrate them. At the Israeli phage therapy centre, EST-003 shares 8 of its patients: 6 with EST-070, another series from the same centre, and the rest with EST-012 and EST-015; a 10-year-old girl with an infected Berlin Heart EXCOR device appears in EST-003, EST-070 and EST-077, and a 7-year-old girl with an osteoarticular infection in EST-003, EST-070 and EST-095. At the University of California San Diego, EST-034 shares 3 patients with EST-049, EST-061 and EST-077. And in the Belgian consortium, EST-108 states that 27 of its 100 patients had been reported before and cites 15 references, of which **7 are studies in this corpus** — EST-002, EST-010, EST-016, EST-046, EST-062, EST-124 and EST-164 — contributing 8 patients; to them is added EST-088, published later, whose patient matches one in EST-108 in infection, species and phages. The examination covered the 65 full texts and gathered 30 pairs: 15 flagged by a detector based on product, country and citation, 8 found by the model's reading of 22 September 2026 and 7 by the authors' re-reading of 30 September. 19 were confirmed, 11 were ruled out and **none remains unread**; the verdict on each pair carries the sentence it rests on and the signature of both authors (annex S21). As the review publishes no pooled estimate, the duplication alters no effect figure, but it prevents reading the corpus as 65 disjoint sets of patients.

5. Conclusions

The clinical evidence base for phage therapy in resistant *P. aeruginosa* is larger than published syntheses reflect and, at the same time, less suited to quantitative synthesis than those syntheses assume. It consists mainly of single cases and non-comparative series, published conditionally on their outcome, with a reporting completeness that does not allow verification of the resistance category defining the population of interest. While those three conditions hold, pooling their outcomes into a common proportion yields a figure that admits no clinical interpretation.

Declarations

Funding. This review received no specific funding from public, commercial or not-for-profit agencies.

Ethics approval and consent. Not applicable. This review synthesises already-published literature and aggregate data from public trial registers; it involves no patients, no specimens and no identifiable individual data, and therefore required neither ethics committee approval nor informed consent. No formal waiver was sought because the institution does not require one for studies without primary data.

Competing interests. The authors declare no competing interests.

Author contributions (CRediT). D. Valdiviezo: conceptualisation, methodology, investigation (according to D.V.'s statement, design of the searches with the participation of the language model and their execution in each interface with the model's guidance; data extraction; risk of bias), direction of the software, data curation, analysis and visualisation, which the language model programmed and ran, review of the drafts, which the model wrote, and project administration. **N. Trelles:** methodology (decision, with D.V., on the outcome rules proposed by the language model, according to D.V.; the record of 12 August 2026 left them as an unratified proposal), investigation, data extraction (122 of the 124 studies), risk of bias and validation. ‹‹N. TRELLES: WRITING — REVIEW AND EDITING, ONLY ONCE CONFIRMED IN WRITING››

Data and code availability. All material supporting this review is kept in a versioned project repository: the search corpus with the raw exports of all nine queries, the append-only decision logs of the three screening stages, both reviewers' extraction workbooks, the methodological decision records with their dates, and the complete pipeline code, which is re-executable from search to the figures in the manuscript. ‹‹PUBLIC DEPOSIT AND DOI — PENDING THE AUTHORS' DECISION››. The decision logs are append-only and preserve every correction alongside the row it supersedes.

Use of artificial intelligence. A generative language model (Claude Opus 5, Anthropic) took part in every phase of this review. **Criteria.** It drafted the proposals for criteria and rules that the authors decided: among them, the study as the unit of inclusion and the eligibility of conference abstracts (5 August 2026), the exclusion codes added after screening (PRO, INT and NOORG) and the outcome rules, which, according to D.V., both authors decided, although the record of 12 August 2026 left them as an unratified proposal; and the two rules added mid-batch on 4 August, which it built into its screening scripts. **Screening.** As sole reviewer and without duplication, it issued the record-by-record decisions of title and abstract screening, including the 35 language exclusions at the abstract stage, and the pre-extraction from abstracts. On the first day of screening it wrote the reasons as free text; afterwards it used six reasons whose wording comes from those texts, and there is no record that the authors fixed them before starting. D.V. was the only human reviewer: according to D.V.'s statement, D.V. examined one by one the reports that survived screening, but that examination was not recorded; those the model excluded were not re-read by any human, except a sample of 350 that D.V. re-screened. **Search.** It took part in designing the equations, which, according to D.V., D.V. ran in each interface with the model's guidance; it drafted the messages of the commits of 3 August 2026, which narrate that execution in the first person, and it downloaded the PubMed and ClinicalTrials.gov records via API. **Selection and extraction.** It located and downloaded full texts, also through a browser; it proposed the grouping of reports into studies; it read the full texts and the registry records and proposed, with their quotation, the exclusions they support, which both authors signed except two signed by D.V. alone; it generated, merged and compared the extraction notebooks; it proposed, with their quotation, a resolution for part of the disagreements of a comparison that was later redone; it extracted the cells of the third patient of EST-121, signed by D.V. alone, and proposed modality corrections that both authors signed; it translated into categories the signed conclusions of the re-reading and applied those that left a single value; and it detected and read first the pairs of studies that could share patients, before the authors' re-reading. **Risk of bias.** It placed in each cell of the forms, as a comment, the sentences of the article it located automatically for each domain, and it generated the consensus notebook with the matching judgements preloaded. **Pipeline and writing.** It wrote and ran all the pipeline code, from which every figure, table and number comes, and it drafted the manuscript, the supplementary files, the cover letter and the decision records. Except for the above, data extraction and risk-of-bias judgements were made by the authors, although the independence of their two readings is not established (sections 2.6 and 2.7). Every decision of the current screening is logged with its reason and timestamp and is supplied in full (S3, S4); the first pass of 4 August over PubMed arm A, which the pool screening replaced, is kept in the repository and not in S3. D.V. takes responsibility for the accuracy and integrity of the work; N.T.'s awaits written approval. No AI tool is listed as an author, in line with ICMJE recommendations.

Tables and figures

The six tables are supplied as separate files (`paper/tablas/`), each with its own footnote. Both figures are supplied as vector PDF (`paper/figuras/`).

Table 1. Characteristics of the retrievable evidence base (n = 65 studies).

Table 2. Reporting completeness for variables critical to stratification.

Table 3. What the full-text criterion leaves out: included studies against those excluded for lacking the article.

Table 4. Exclusion reasons by stage, using the closed vocabulary.

Table 5. Reporting completeness for the five declared outcomes, over the adjudicated extraction (n = 73 arms).

Table 6. Arms surviving each requirement for a pooled proportion of clinical success (n = 73 arms).

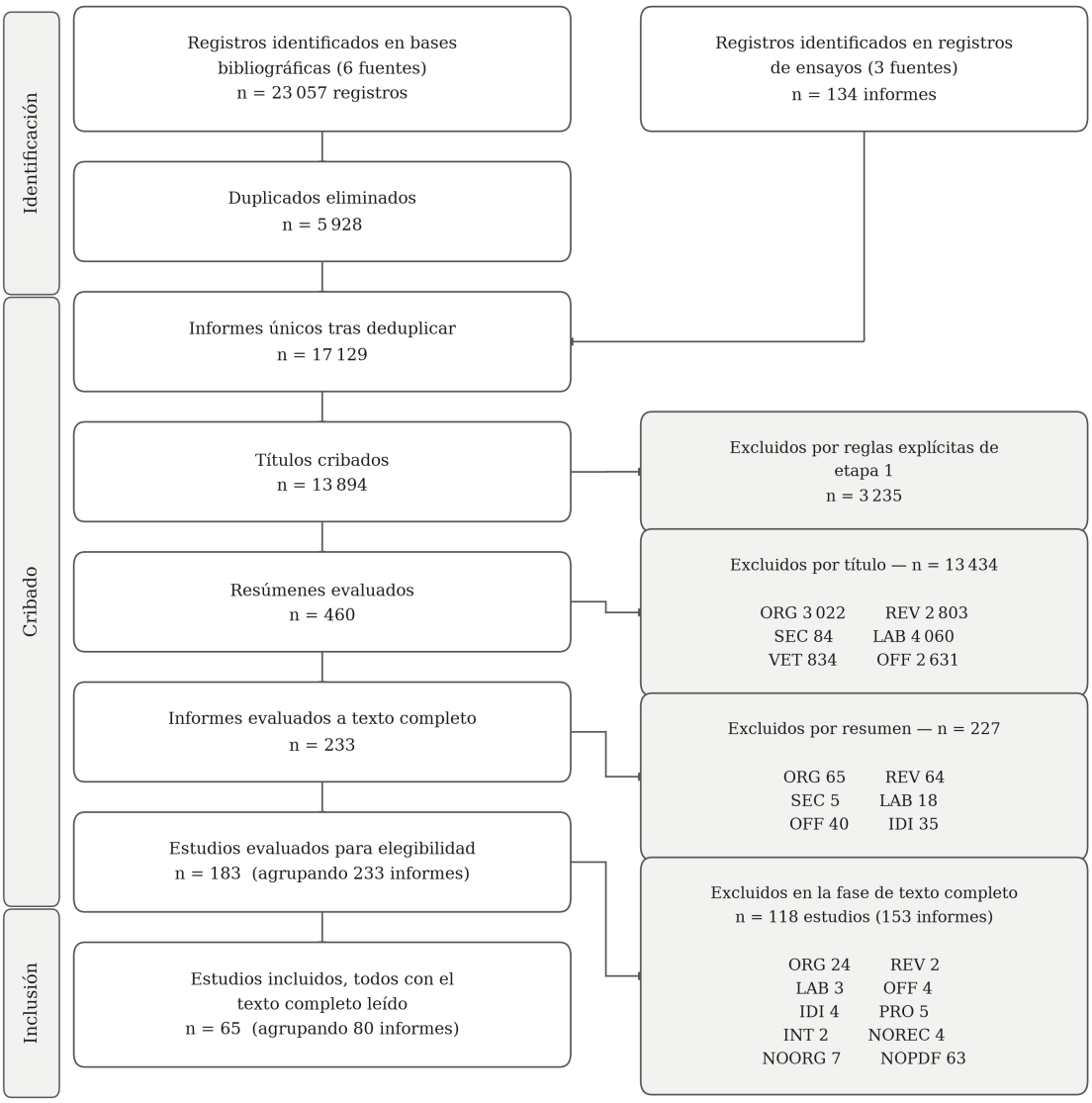


Figure 1. PRISMA 2020 flow diagram.

What it shows. The passage from identified records to included studies, with the two identification streams — bibliographic databases and trial registers — counted separately, as the PRISMA 2020 statement requires. *How to read it.* The left column is the main flow; the grey boxes on the right hold what was excluded at each step, broken down by the closed six-code vocabulary (ORG, REV, SEC, LAB, VET, OFF; defined in Table 4). The register stream enters from the side and does not pass through bibliographic deduplication, because a registry entry has no DOI with which to collide against an article. The final box holds the included studies, all with their full text read, which is the entry condition since the criterion of section 2.9. *Source.* Project screening pipeline; documented search runs of 3 and 4 August 2026, with the BVS export added to the corpus on 10 August.

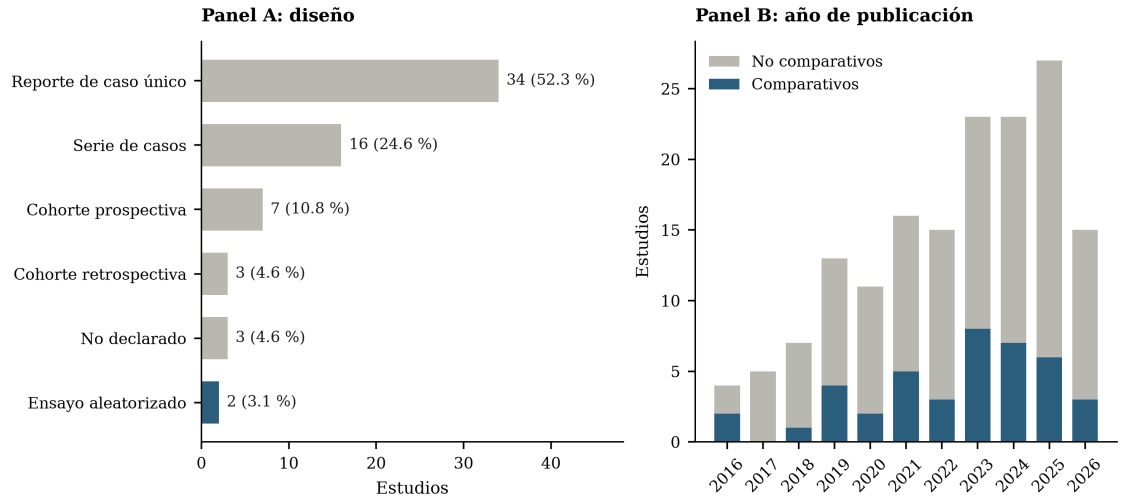


Figure 2. Composition of the retrievable evidence base (n = 65 studies).

What it shows. Panel A orders studies by design; Panel B distributes them by year of publication, separating comparative designs from the rest within each year. *How to read it.* In both panels the dark colour marks comparative designs — randomised and non-randomised trials — the only ones able to sustain a claim of relative efficacy. "Not stated" in Panel A means the abstract does not allow the design to be recognised, not that the study lacks one. The year 2026 is incomplete: the search closed in August. *Source.* Systematic pre-extraction from the abstracts of the 95 studies with a retrievable publication.

Supplementary material

- **S0.** Pre-review assessment of the manuscript drafted by the language model in the role of a journal editor; it is not an external editorial assessment.
- **S1.** PRISMA 2020 checklist.
- **S2.** Verbatim search strings by source, with their number of results.
- **S3.** The decisions of the screening in force, in two files: stage 2 (title) and stage 3 (abstract), each with its reason, who issued it and its timestamp. The first pass of 4 August 2026 over arm A of PubMed, which the pool screening replaced, is kept in the repository and not in S3.
- **S4.** Pre-extraction from the abstracts of the 65 studies with a retrievable publication, issued by the language model and flagged as partial in every record.
- **S5.** Listing of the 65 studies in the corpus, with their status and grouped reports. The 118 excluded after screening are in S16.
- **S6.** Audit of the screening against the 42 positive controls fixed before it began, 40 of them checkable by PMID.
- **S7.** ICMJE declarations.
- **S8.** Retrieval route resolved in August 2026 for each report that went to full text without an editorial DOI, PMID or NCT: the registry identifier or protocol code, the DOI located, the already identified trial it reports on, or the citation to search by. It does not record the download attempts or their outcome; the studies left without the full article are in S16 (NOPDF).
- **S9.** Language by report, with the class of evidence it provided.
- **S10.** Language of the body of the PDF of each of the 65 studies in the corpus: a mechanical detection over the text of the whole PDF, not over the metadata nor over the abstract alone, made by a pipeline script written by the language model, introduced on 6 October 2026 and redone on every run. It gives 65 in

English; the mirar_a_mano ('check by hand') column flags 1 PDF for page-by-page review because of its share of Cyrillic or CJK characters.

- **S11.** Agreement between the two extractions, whose independence is not established, before disagreements were resolved.
- **S12.** How the 541 extraction disagreements were closed and who adjudicated them according to the log: both reviewers in 14 and a single reviewer in 525; 2 were closed by a mechanical rule, without joint sign-off.
- **S13.** Outcome extraction rules, with their documented correction. The language model drafted them; according to D.V., both authors decided them, and the record of 12 August 2026 left them as an unratified proposal.
- **S14.** Adjudicated extraction: one value per cell, arm by arm, after the disagreements were adjudicated. Underpins section 3.5 and Tables 5 and 6.
- **S15.** Provenance of every cell in S14: agreement, adjudicated by both reviewers, adjudicated by one, corrected against the text, single reading, or open.
- **S16.** The 118 studies the screening admitted that were excluded afterwards, with their reason code and who signed the exclusion. 47 carry the quotation that supports it; the 63 without the full article (NOPDF) and the 4 without readable text (NOREC) have nothing to quote, and 4 more carry no quotation. Of those with a quotation, 17 quote the registry record and 1 the SciELO indexing.
- **S17.** Table 6 broken down: the seven requirements, arm by arm over the 73 arms, with the first one that fails and why.
- **S18.** The 68 risk-of-bias judgements, one per row, with what supports each in the signed notebook: a verbatim quotation in 24, a note without a quotation in 36 and nothing in 8, flagged as missing data.
- **S18b.** The sentences from each article that the language model located automatically for each risk-of-bias domain and placed as comments in the forms; they accompany what supports each judgement in S18, they do not replace it.
- **S19.** Which model issued the screening decisions, when, over which files, what D.V. examined, according to D.V.'s statement, and what was not recorded, flagged as missing data.
- **S20.** Which report each of the 73 arms came from, and with what evidence the link was established.
- **S21.** The 30 pairs of studies that could describe the same patients, with the signal that flagged them, the verdict after reading both articles, the patients shared and the signed sentence it rests on.
- **S22.** For each stratification variable, how many studies state it in the full text, how many mention it without allowing classification, how many are silent, and how many have no text at all.
- **S23.** The resistance class of the 70 studies that had full text, checked against the article's antibiogram: the signed conclusion, the sentence or table it rests on, how it was coded and what changed in the extraction.
- **S24.** The 14 studies with a comparative design, re-extracted: comparator, allocation, time zero, assessment time, co-interventions, losses, estimand and whether there is a contrast for *P. aeruginosa*.

A guide accompanies them explaining, file by file, what each contains and what question it lets the reader answer.

References

1. Tacconelli E, Carrara E, Savoldi A, Harbarth S, Mendelson M, Monnet DL, et al. Discovery, Research, and Development of New Antibiotics: The WHO Priority List of Antibiotic-Resistant Bacteria and Tuberculosis. *Lancet Infectious Diseases*. 2018;18(3):318-327. doi:10.1016/S1473-3099(17)30753-3

2. World Health Organization. WHO Bacterial Priority Pathogens List, 2024: Bacterial Pathogens of Public Health Importance to Guide Research, Development and Strategies to Prevent and Control Antimicrobial Resistance. 2024.

3. Pirnay J, Djebara S, Steurs G, Griselain J, Cochez C, De Soir S, et al. Personalized Bacteriophage Therapy Outcomes for 100 Consecutive Cases: A Multicentre, Multinational, Retrospective Observational Study. *Nature Microbiology*. 2024;9(6):1434-1453. doi:10.1038/s41564-024-01705-x

4. Aslam S, Courtwright AM, Koval C, Lehman SM, Morales S, Furr C, et al. Early Clinical Experience of Bacteriophage Therapy in 3 Lung Transplant Recipients. *American Journal of Transplantation*. 2019;19(9):2631-2639. doi:10.1111/ajt.15503

5. Chan BK, Turner PE, Kim S, Mojibian HR, Elefteriades JA, Narayan D. Phage Treatment of an Aortic Graft Infected With Pseudomonas aeruginosa. *Evolution, Medicine, and Public Health*. 2018;2018(1):60-66. doi:10.1093/emph/eoy005

6. Uyttebroek S, Chen B, Onsea J, Ruythooren F, Debaveye Y, Devolder D, et al. Safety and Efficacy of Phage Therapy in Difficult-to-Treat Infections: A Systematic Review. *Lancet Infectious Diseases*. 2022;22(8):e208-e220. doi:10.1016/S1473-3099(21)00612-5

7. Liu Y, Thong S, Moreira W, Yeo JH, Zhong Y, Chong ZS, et al. Clinical Application of Customized and Non-Customized Bacteriophage Therapy in Patients with Refractory/Resistant Bacterial Infections: A Systematic Review and Meta-Analysis. *International Journal of Antimicrobial Agents*. 2025;66(4):107570. doi:10.1016/j.ijantimicag.2025.107570

8. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 Statement: An Updated Guideline for Reporting Systematic Reviews. *BMJ*. 2021;372:n71. doi:10.1136/bmj.n71

9. Magiorakos A, Srinivasan A, Carey RB, Carmeli Y, Falagas ME, Giske CG, et al. Multidrug-Resistant, Extensively Drug-Resistant and Pandrug-Resistant Bacteria: An International Expert Proposal for Interim Standard Definitions for Acquired Resistance. *Clinical Microbiology and Infection*. 2012;18(3):268-281. doi:10.1111/j.1469-0691.2011.03570.x

10. Sterne JAC, Savovi\'c J, Page MJ, Elbers RG, Blencowe NS, Boutron I, et al. RoB 2: A Revised Tool for Assessing Risk of Bias in Randomised Trials. *BMJ*. 2019;366:l4898. doi:10.1136/bmj.l4898

11. Sterne JA, Hernan MA, Reeves BC, Savovi\'c J, Berkman ND, Viswanathan M, et al. ROBINS-I: A Tool for Assessing Risk of Bias in Non-Randomised Studies of Interventions. *BMJ*. 2016;355:i4919. doi:10.1136/bmj.i4919

12. Murad MH, Sultan S, Haffar S, Bazerbachi F. Methodological Quality and Synthesis of Case Series and Case Reports. *BMJ Evidence-Based Medicine*. 2018;23(2):60-63. doi:10.1136/bmjebm-2017-110853

13. Munn Z, Barker TH, Moola S, Tufanaru C, Stern C, McArthur A, et al. Methodological Quality of Case Series Studies: An Introduction to the JBI Critical Appraisal Tool. *JBI Evidence Synthesis*. 2020;18(10):2127-2133. doi:10.11124/JBISRIR-D-19-00099

14. Guyatt G, Oxman AD, Akl EA, Kunz R, Vist G, Brozek J, et al. GRADE Guidelines: 1. Introduction --- GRADE Evidence Profiles and Summary of Findings Tables. *Journal of Clinical Epidemiology*. 2011;64(4):383-394. doi:10.1016/j.jclinepi.2010.04.026

15. Barker TH, Migliavaca CB, Stein C, Colpani V, Falavigna M, Aromataris E, et al. Conducting Proportional Meta-Analysis in Different Types of Systematic Reviews: A Guide for Synthesisers of Evidence. *BMC Medical Research Methodology*. 2021;21(1):189. doi:10.1186/s12874-021-01381-z

16. Migliavaca CB, Stein C, Colpani V, Barker TH, Ziegelmann PK, Munn Z, et al. Meta-Analysis of Prevalence: I^2 Statistic and How to Deal With Heterogeneity. *Research Synthesis Methods*. 2022;13(3):363-367. doi:10.1002/jrsm.1547

17. Jault P, Leclerc T, Jennes S, Pirnay J, Que Y, Resch G, et al. Efficacy and Tolerability of a Cocktail of Bacteriophages to Treat Burn Wounds Infected by Pseudomonas aeruginosa (PhagoBurn): A Randomised, Controlled, Double-Blind Phase 1/2 Trial. *Lancet Infectious Diseases*. 2019;19(1):35-45. doi:10.1016/S1473-3099(18)30482-1

18. Weiner I, Kahan-Hanum M, Buchstab N, Zelcbuch L, Navok S, Sherman I, et al. Phage Therapy with Nebulized Cocktail BX004-A for Chronic Pseudomonas aeruginosa Infections in Cystic Fibrosis: A Randomized First-in-Human Trial. *Nature Communications*. 2025;16(1):5579. doi:10.1038/s41467-025-60598-4

19. Nir-Paz R, Onallah H, Dekel M, Gellman Y, Haze A, Ben-Ami R, et al. Randomized double-blind study on safety and tolerability of TP-102 phage cocktail in patients with infected and non-infected diabetic foot ulcers. *Med*. 2025;6:100565. doi:10.1016/j.medj.2024.11.018

20. Ferry T, Kolenda C, Laurent F, Leboucher G, Merabishvili M, Djebara S, et al. Personalized Bacteriophage Therapy to Treat Pandrug-Resistant Spinal Pseudomonas aeruginosa Infection. *Nature Communications*. 2022;13:4239. doi:10.1038/s41467-022-31837-9

21. Onallah H, Hazan R, Nir-Paz R, others. Refractory Pseudomonas aeruginosa Infections Treated with Phage PASA16: A Compassionate Use Case Series. *Med*. 2023;4(9):600-611.e4. doi:10.1016/j.medj.2023.07.002